Symmetric tensor categories

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Lecture 1

The model example: $\mathrm{Rep}_{\Bbbk}(G)$

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We would now like to axiomatize the categorical structures visible in $\mathrm{Rep}_{\Bbbk}(G)$, without referring to underlying vector spaces.

Monoidal category: data

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with the following properties.

Monoidal category: pentagon axiom

First, the associativity constraint is required to satisfy the **pentagon axiom**: for all $W, X, Y, Z \in \mathcal{C}$, the diagram

$$\begin{tikzcd}
((W \otimes X) \otimes Y) \otimes Z \arrow[rr, "a_{W \otimes X, Y, Z}"] \arrow[d, "a_{W,X,Y} \otimes \mathrm{id}_Z"'] & & (W \otimes X) \otimes (Y \otimes Z) \arrow[d, "a_{W,X,Y \otimes Z}"] \\
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commutes. Equivalently,

$$a_{W,X,Y \otimes Z}\, a_{W \otimes X,Y,Z} = (\mathrm{id}_W \otimes a_{X,Y,Z})\, a_{W,X \otimes Y,Z}\, (a_{W,X,Y} \otimes \mathrm{id}_Z).$$

Monoidal category: unit axiom

Next, the unit object $\mathbf{1}$ is required to satisfy

$$\mathbf{1} \otimes \mathbf{1} \cong \mathbf{1},$$

and tensor multiplication by $\mathbf{1}$ on either side must be an equivalence:

$$L_{\mathbf{1}} : \mathcal{C} \to \mathcal{C}, \qquad X \mapsto \mathbf{1} \otimes X,$$

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The **Maclane's coherence theorem** implies that we may ignore parentheses and unit objects inside tensor products.

Braided monoidal category: data

A **braided monoidal category** is a monoidal category $\mathcal{C}$ together with a natural isomorphism

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They imply that for every object $X \in \mathcal{C}$ the object $X^{\otimes n}$ carries a natural action of the **braid group** B_n, with generator b_i acting by c in the i-th and $i+1$-th factors.

Braided monoidal category: hexagon axioms

1. For all $X, Y, Z \in \mathcal{C}$, the diagram

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\begin{tikzcd}
(X \otimes Y) \otimes Z \arrow[r, "a_{X,Y,Z}"] \arrow[d, "c_{X,Y} \otimes \mathrm{id}_Z"'] & X \otimes (Y \otimes Z) \arrow[r, "c_{X,Y \otimes Z}"] & (Y \otimes Z) \otimes X \arrow[d, "a_{Y,Z,X}"] \\
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X \otimes (Z \otimes Y) \arrow[r, "a^{-1}_{X,Z,Y}"'] & (X \otimes Z) \otimes Y \arrow[r, "c_{X,Z} \otimes \mathrm{id}_Y"'] & (Z \otimes X) \otimes Y \arrow[u, "a_{Z,X,Y}"']
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Symmetric monoidal category

A braided monoidal category is **symmetric** if the braiding is involutive:

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In $\mathrm{Rep}_{\mathbb{k}}(G)$, c_{VW} is the flip

$$v \otimes w \mapsto w \otimes v.$$

Rigidity: dual data

Let $\mathcal{C}$ be monoidal. A **left dual** of $X \in \mathcal{C}$ is an object X^* with morphisms

$$\mathrm{ev}_X : X^* \otimes X \to \mathbf{1}, \qquad \mathrm{coev}_X : \mathbf{1} \to X \otimes X^*,$$

satisfying the two triangular identities.

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A **right dual** is defined analogously, with evaluation $X \otimes {}^*X \to \mathbf{1}$ and coevaluation $\mathbf{1} \to {}^*X \otimes X$. A monoidal category is **rigid** if every object has both a left and a right dual.

Rigidity: triangular identities

The left-dual triangular identities are the commutativity of the following two diagrams:

$$\begin{tikzcd} X \arrow[r, "\mathrm{coev}_X \otimes \mathrm{id}_X"] \arrow[rrr, bend right, "\mathrm{id}_X"'] & (X \otimes X^*) \otimes X \arrow[r, "a_{X,X^*,X}"] & X \otimes (X^* \otimes X) \arrow[r, "\mathrm{id}_X \otimes \mathrm{ev}_X"] & X \end{tikzcd}$$

$$\begin{tikzcd} X^* \arrow[r, "\mathrm{id}_{X^*} \otimes \mathrm{coev}_X"] \arrow[rrr, bend right, "\mathrm{id}_{X^*}"'] & X^* \otimes (X \otimes X^*) \arrow[r, "a^{-1}_{X^*,X,X^*}"] & (X^* \otimes X) \otimes X^* \arrow[r, "\mathrm{ev}_X \otimes \mathrm{id}_{X^*}"] & X^* \end{tikzcd}$$

Equivalently, the corresponding composites are id_X and id_{X^*}.

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Note that if X^* or ${}^*X$ exists for a given X then it is unique up to a unique isomorphism, and that in a symmetric category we have a canonical isomorphism ${}^*X \cong X^*$, so there is only one dual.

Rigidity in $\mathrm{Rep}_{\Bbbk}(G)$

For $V \in \mathrm{Rep}_{\Bbbk}(G)$, set

$$V^* = \mathrm{Hom}_{\Bbbk}(V, \Bbbk),$$

with contragredient G-action. Evaluation and coevaluation are

$$\mathrm{ev} : V^* \otimes V \to \Bbbk, \qquad \varphi \otimes v \mapsto \varphi(v),$$

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for a basis $\{v_i\}$ and its dual basis $\{v_i^*\}$. These maps are G-equivariant and satisfy the triangular identities.

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A braided tensor category is called **symmetric** is its braiding is symmetric.

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It is a whole world where we can do algebra and representation theory!

Examples of symmetric tensor categories

For example, $\mathrm{Rep}_{\Bbbk}(G)$ is a symmetric tensor category. The most basic case is $G = 1$, which gives **the category of finite dimensional $\Bbbk$-vector spaces**, $\mathrm{Vec}_{\Bbbk}$.

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A **rational representation** of an affine group scheme G is a finite-dimensional $\mathcal{O}(G)$-**comodule** V, i.e., a vector space V with a **coaction**

$$\rho : V \to V \otimes \mathcal{O}(G)$$

which defines an action of the algebra $\mathcal{O}(G)^*$ on V. It is easy to see that rational representations of an affine group scheme G form a symmetric tensor category $\mathrm{Rep}(G)$.

Tensor functors

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$$J_{X,Y} : F(X) \otimes F(Y) \xrightarrow{\sim} F(X \otimes Y), \qquad \nu : \mathbf{1}_{\mathcal{D}} \xrightarrow{\sim} F(\mathbf{1}_{\mathcal{C}}),$$

compatible with the associativity and unit data. A **braided (symmetric) monoidal functor** is a monoidal functor between braided (symmetric) monoidal categories compatible with the braidings.

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If $\mathcal{C}, \mathcal{D}$ are tensor categories then a monoidal functor $F : \mathcal{C} \to \mathcal{D}$ is called a **tensor functor** if it is $\mathbb{k}$-linear and faithful (hence exact). A **fiber functor** is a symmetric tensor functor to the target $\mathrm{Vec}_{\mathbb{k}}$.

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Thus we would like to explain why the previous examples **are equivalent** to $\mathrm{Rep}(G)$ for some affine group scheme G as symmetric tensor categories.

Fiber functors in the examples

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This is a **fiber functor**: a symmetric tensor functor to $\mathrm{Vec}_{\Bbbk}$.

Tensor automorphisms of a fiber functor

Let $\omega : \mathcal{C} \to \mathrm{Vec}_{\Bbbk}$ be a fiber functor. Define the affine group functor

$$\underline{\mathrm{Aut}}_{\otimes}(\omega) : R \mapsto \mathrm{Aut}^{\otimes}(\omega_R), \qquad \omega_R(X) = \omega(X) \otimes_{\Bbbk} R.$$

Thus an R-point of $\underline{\mathrm{Aut}}_{\otimes}(\omega)$ is a family

$$\{g_X : \omega(X) \otimes R \xrightarrow{\sim} \omega(X) \otimes R\}_{X \in \mathcal{C}}$$

which is natural in X, preserves tensor products and the unit, and is invertible.

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For $\mathcal{C} = \mathrm{Rep}(G)$ and $\omega = \omega_G$, evaluation of representations gives

$$G \cong \underline{\mathrm{Aut}}_{\otimes}(\omega_G).$$

The Coend of matrix coefficients

The functor $\underline{\mathrm{Aut}}_{\otimes}(\omega)$ is represented by a Hopf algebra constructed from matrix coefficients. Set

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Concretely, this is the quotient of

$$\bigoplus_{X\in\mathcal{C}} \omega(X)^* \otimes \omega(X)$$

by the relations

$$\omega(f)^*\varphi \otimes v - \varphi \otimes \omega(f)v \in \omega(Y)^* \otimes \omega(Y)$$

for all morphisms $f : X \to Y$, where $\varphi \in \omega(Y)^*$ and $v \in \omega(X)$.

Hopf algebra structure on the Coend

Write the class of $\psi \otimes v \in \omega(X)^* \otimes \omega(X)$ as $[\psi \mid v]_X$. The coalgebra structure is the usual one for matrix coefficients:

$$\Delta([\psi \mid v]_X) = \sum_i [\psi \mid e_i]_X \otimes [e_i^* \mid v]_X, \qquad \varepsilon([\psi \mid v]_X) = \psi(v),$$

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The tensor structure of ω defines multiplication:

$$[\psi \mid v]_X \, [\varphi \mid w]_Y = [\psi \otimes \varphi \mid v \otimes w]_{X \otimes Y},$$

after identifying $\omega(X \otimes Y) \cong \omega(X) \otimes \omega(Y)$. Duals give the antipode.

Tannakian reconstruction

Let $\mathcal{C}$ be a symmetric tensor category and let

$$\omega : \mathcal{C} \to \mathrm{Vec}_{\Bbbk}$$

be a fiber functor. Put

$$G := \underline{\mathrm{Aut}}_{\otimes}(\omega), \qquad \mathcal{O}(G) = \int^{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X).$$

Then the canonical coactions on $\omega(X)$ define a symmetric tensor equivalence

$$\mathcal{C} \simeq \mathrm{Rep}(G).$$

This shows that all our examples are of the form $\mathrm{Rep}(G)$ for an affine group scheme G.

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It was shown by Deligne and Milne that the fiber functor is unique up to isomorphism; consequently G is uniquely determined up to conjugation.

Abstract groups and proalgebraic completion

Let us now explain how this works in our first example. Let Γ be an abstract group. The forgetful functor gives

$$\widehat{\Gamma} := \underline{\mathrm{Aut}}_{\otimes}(\omega_{\Gamma}), \qquad \mathrm{Rep}_{\Bbbk}(\Gamma) \simeq \mathrm{Rep}(\widehat{\Gamma}).$$

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Its coordinate algebra is the subalgebra of $\mathrm{Fun}(\Gamma, \Bbbk)$ spanned by the matrix coefficients

$$g \mapsto \langle f, gv \rangle, \qquad v \in V, \quad f \in V^*, \quad V \in \mathrm{Rep}_{\Bbbk}(\Gamma).$$

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Thus the group scheme $\widehat{\Gamma}$ is reduced, i.e., it is a proalgebraic group. Namely, it is the inverse limit of algebraic groups G equipped with a homomorphism from Γ, i.e.,

$$\widehat{\Gamma} = \varprojlim_{\Gamma \to G} G.$$

In particular, $\widehat{\Gamma}$ carries a canonical (universal) homomorphism $\Gamma \to \widehat{\Gamma}$ (which may or may not be injective).

A symmetric tensor category $\mathcal{C}$ over $\Bbbk$ is called **Tannakian** if it admits a fiber functor

$$\omega : \mathcal{C} \to \mathrm{Vec}_{\Bbbk} \, .$$

Equivalently, by reconstruction,

$$\mathcal{C} \simeq \mathrm{Rep}(G)$$

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Thus Tannakian categories are precisely the symmetric tensor categories whose objects can be realized as finite-dimensional vector spaces, functorially and compatibly with tensor products, duals, and the symmetry.

So it is natural to ask: **are there other examples of symmetric tensor categories**?

The first non-Tannakian example - supervector spaces

The answer turns out to be **yes**. Assume $\operatorname{char} \Bbbk \neq 2$. Let $\mathrm{sVec}_{\Bbbk}$ be the category of finite-dimensional **supervector spaces**

$$V = V_{\bar{0}} \oplus V_{\bar{1}}.$$

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The underlying tensor category is equivalent to $\mathrm{Rep}_{\Bbbk}(\mathbb{Z}/2)$, but the symmetric structure is different.

But why is it a genuinely new example?

Trace and categorical dimension

Let X be an object of a symmetric tensor category and let $f : X \to X$. The **categorical trace** is

$$\mathrm{Tr}(f) : \mathbf{1} \xrightarrow{\mathrm{coev}_X} X \otimes X^* \xrightarrow{f \otimes 1} X \otimes X^* \xrightarrow{c_{X,X^*}} X^* \otimes X \xrightarrow{\mathrm{ev}_X} \mathbf{1}.$$

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In $\mathrm{Vec}_\Bbbk$, the trace is the usual trace and dimension is the usual dimension viewed as an element of $\Bbbk$. In $\mathrm{sVec}_\Bbbk$, the trace is the **supertrace** $\mathrm{Tr}(f) = \mathrm{Tr}(f_0) - \mathrm{Tr}(f_1)$, and the dimension is the **superdimension**

$$\dim(V) = \dim_\Bbbk V_{\bar{0}} - \dim_\Bbbk V_{\bar{1}}.$$

Invertible objects distinguish sVec from Tannakian categories

An object X in a monoidal category is **invertible** if there exists Y such that

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How can we generalize this example?

Affine supergroup schemes

Affine supergroup schemes

An **affine superscheme** X over $\Bbbk$ is represented by a commutative algebra $\mathcal{O}(X) = \mathcal{O}(X)_{\bar{0}} \oplus \mathcal{O}(X)_{\bar{1}}$ in $\mathrm{sVec}_{\Bbbk}$ (a.k.a. **supercommutative algebra**).

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Equivalently, G is a representable group-valued functor of points on supercommutative $\Bbbk$-algebras.

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Basic examples of affine supergroup schemes are classical algebraic supergroups, such as $GL(m|n)$ and $Osp(m|2n)$.

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Basic examples of affine supergroup schemes are classical algebraic supergroups, such as $GL(m|n)$ and $Osp(m|2n)$.

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There is a slight generalization of this example. Suppose $z \in G(\Bbbk)$ is an element of order 2 acting on $\mathcal{O}(G)$ by parity. Then we may consider the category $\mathcal{C} \simeq \mathrm{Rep}(G, z)$ of representations of G on which z acts by parity. This is also a symmetric tensor category, and $\mathrm{Rep}(G) \cong \mathrm{Rep}(\{1, z\} \ltimes G, z)$, so it generalizes the previous example.

Superfiber functors and super-Tannakian categories

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Super-Tannakian reconstruction

Let

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be a superfiber functor. Deligne showed that in this case $\mathcal{C} = \mathrm{Rep}(G, z)$ for a unique pair (G, z) up to conjugation.

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The coordinate Hopf superalgebra is again a Coend of matrix coefficients:

$$\mathcal{O}(G) = \int^{X \in \mathcal{C}} \omega(X)^* \otimes \omega(X),$$

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Since ω is unique, so is (G, z).

Are there examples beyond super-Tannakian categories?

We have thus enlarged the Tannakian class from fiber functors to $\mathrm{Vec}_{\Bbbk}$ to fiber functors to $\mathrm{sVec}_{\Bbbk}$, i.e., to super-Tannakian categories. The next question is:

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However, the answer in characteristic zero becomes **yes** under the condition of **moderate growth**.

Moderate growth

Let $\mathcal{C}$ be a tensor category. For $X \in \mathcal{C}$, let

$$\ell_n(X) := \ell(X^{\otimes n})$$

be the length of the n-th tensor power. The category $\mathcal{C}$ has **moderate growth** if for every X there exists $C_X > 0$ such that

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Thus every super-Tannakian category has moderate growth.

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Equivalently, under moderate growth there exists a superfiber functor

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This shows that while we can do algebra in any symmetric tensor category, in characteristic zero and moderate growth setting such algebra can be described in classical terms, as ordinary equivariant algebra under a supergroup scheme.